

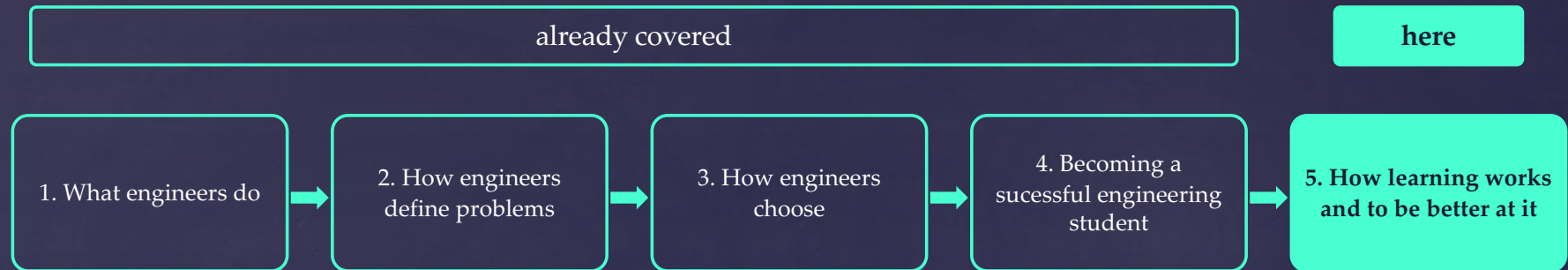
ENGR 111

**Introduction to Engineering
Class #7**

How people learn

How learning works, so you can get better at it

Where we are today



Questions we will explore here.

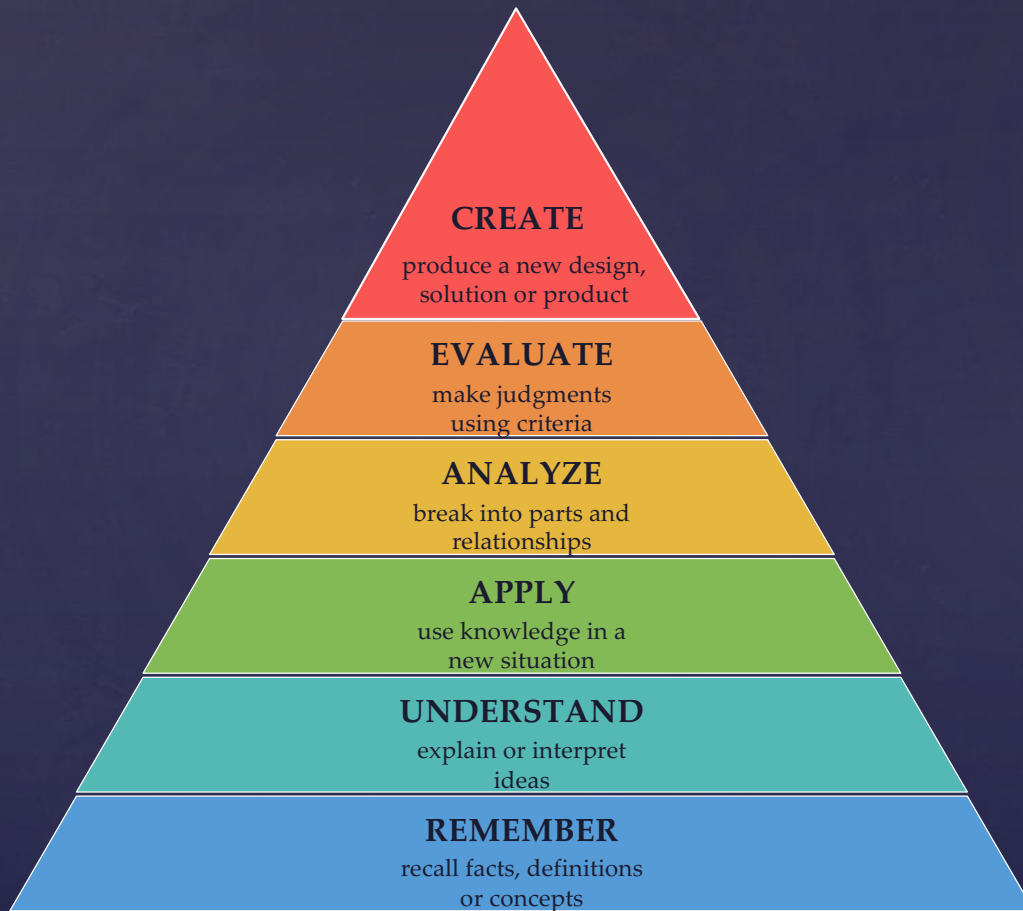
- How do we learn?
- How deeply are we engaging with and using what we learn?
- What makes learning stick after the lecture is over?

Why we are covering this?

- Engineering courses don't teach "how to learn", but it forms the basis of your entire engineering education.

Bloom's taxonomy

Framework for categorizing educational goals (A way to name how deeply you can use what you learned).



The idea. Learning can mean more than remembering a fact. It can mean explaining it, using it, taking it apart, judging it or making something new.

Why it matters. A course can ask for any level. The verb usually tells you which one: list, explain, solve, compare, defend, design.

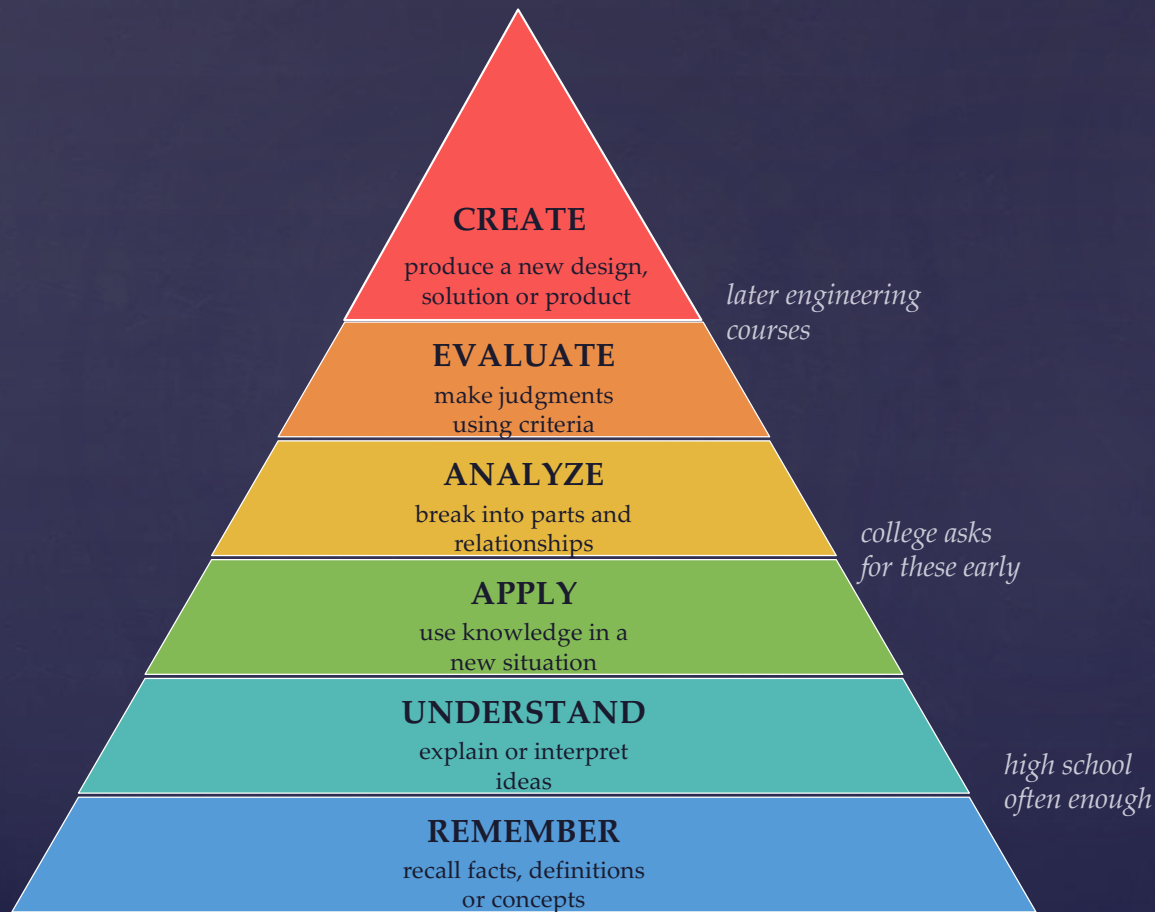
The version shown here. The 1956 taxonomy was revised in 2001. This is the common revised pyramid.

The question is not just what you know. It is what you can do with it.

Suppose you listen to a one hour podcast on the strangeness of quantum mechanics, what level on the pyramid is your learning you think?

80/20 Learning Rule

Different from the Pareto rule: this is the textbook's high-school/college rule of thumb.



High school. About 80 percent of what you needed came from the teacher and class time.

College reverses it. About 20 percent comes from lecture. The other 80 percent is what you do outside class.

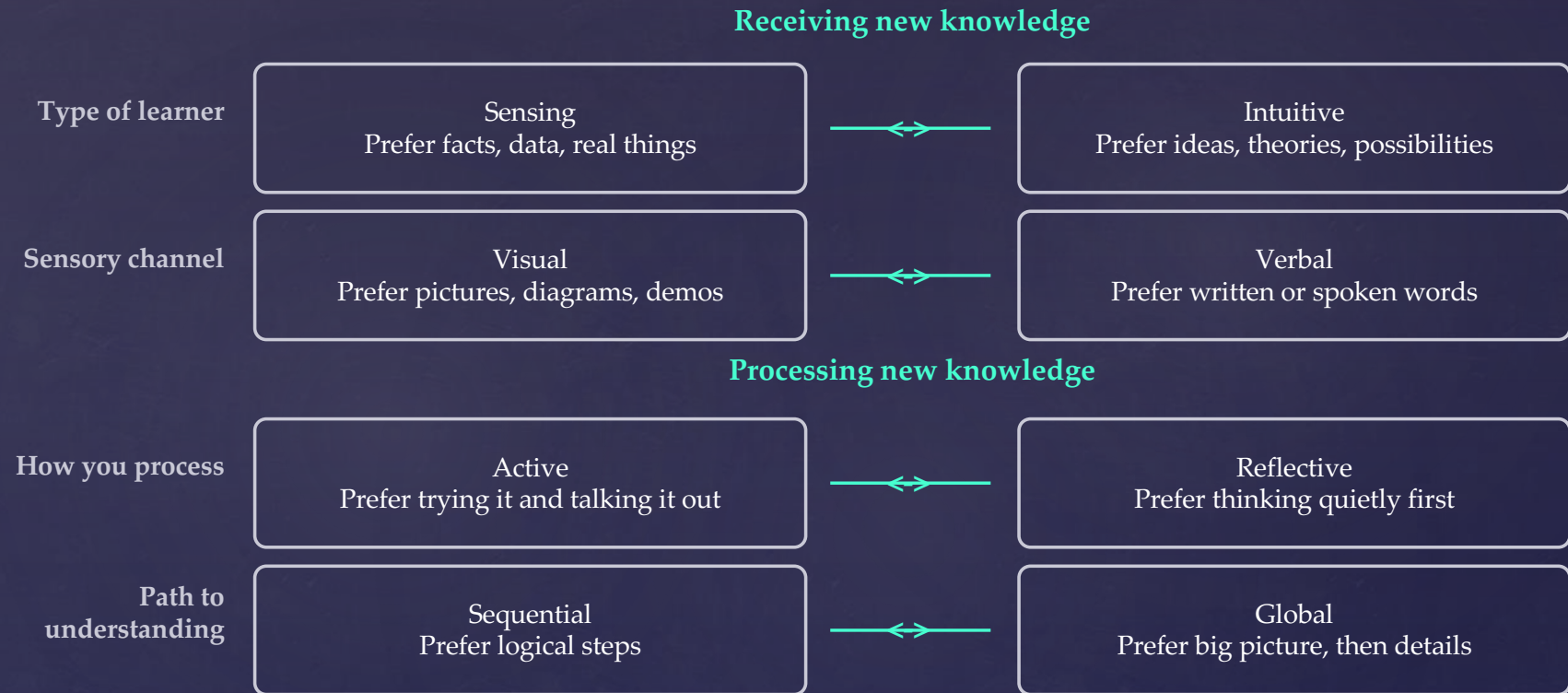
This connects to Bloom. Lecture can support remembering and understanding. Applying and analyzing need practice after class.

The learning comes from use. You learn it when you work problems, check answers, explain the method and try again.

What you do after class in college is not extra. It is most of the course and where most of the learning comes from.

Four learning-style scales

Learning styles: How do you prefer to receive and process new knowledge?



The key word is "prefer". This is about the route that feels natural first, not the only route you can use.

How do you measure on all four scales?

Which side of each scale does a traditional engineering lecture feed?

Use the scales as a checklist

The Index of Learning Styles gives a score on each of the four scales.

<https://learningstyles.webtools.ncsu.edu>

The questionnaire. Forty-four choices. It reports whether you are strong, moderate or balanced on each scale.

Support your preferred learning styles. It can help you compensate when the class is taught in a way that does not fit you.

Use the opposite side too. It gives another perspective on the material and builds skills you would otherwise skip.

The textbook's boxed line. All learners benefit from using both sides of all four dimensions.

Learning is reinforcement

The textbook's principle: repeated exposure over time is how material sticks.

Repeated exposure. You do not learn mechanics, calculus or chemistry in one pass. You meet the ideas again and again.

Each pass is different. Read it, hear it, work it, explain it, correct it, use it later in another course.

This is not just memory. Reinforcement is how a method becomes available when nobody tells you which method to use.

Cramming has a limit. It can get you through an exam. It does not give long-term retention for the next course.

Spaced review changes the curve

Review soon, then again later, before the material has to be relearned.

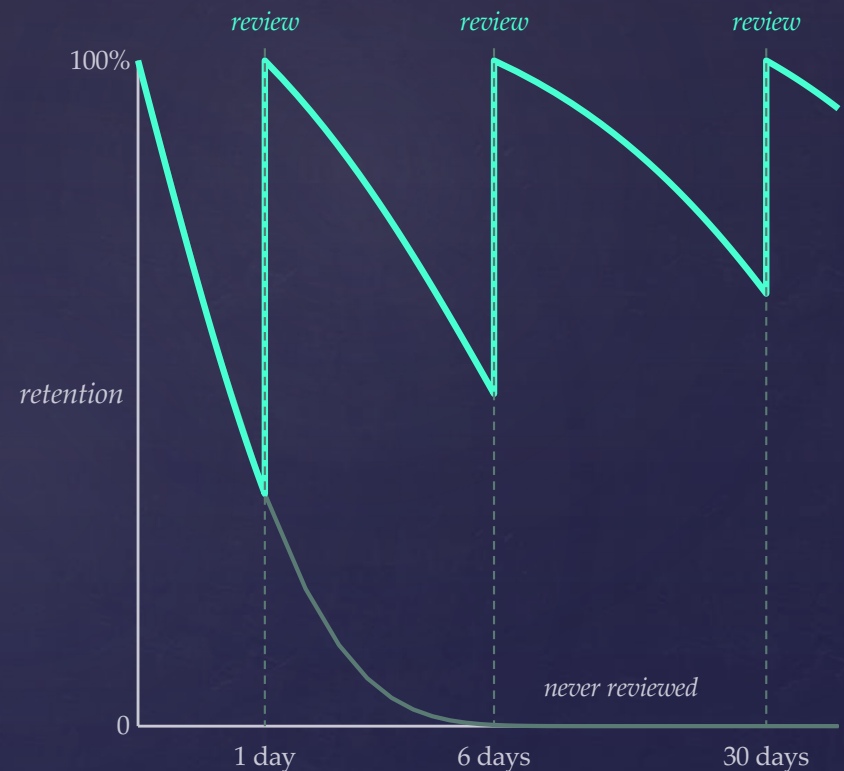
After one day. The textbook says you can forget over 60 percent of what you studied.

Through review. You can regain close to 100 percent, then retain the material longer.

The reviews get shorter. About ten minutes within a day, five minutes around six days, and two to four minutes around thirty days.

Use this to prepare for exams. Review in multiple short sessions before an exam (not just in one cram session the night before).

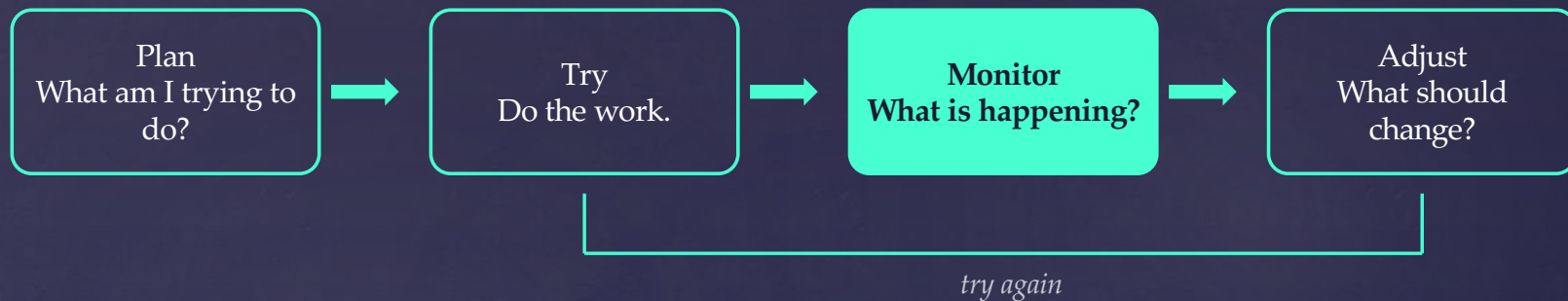
Skip reviews. The exam push becomes relearning, and the next course gets harder.



Schedule the first review while the lecture is still fresh.

Metacognition

A feedback loop: plan, try, monitor, adjust, then try again.



Situation	Monitor	Adjust
Problem set	I cannot start problem 5	Go get a hint, then restart cold
Studying	I keep rereading the same page	Close the notes and test myself
Lecture	I am lost right now	Ask, or mark the exact question

The textbook's three words. Plan, monitor, evaluate.

The expert learner's habit. Notice what is not working, then change something while there is still time.

If the same approach keeps giving the same result, change the approach.

Before, during, and after a lecture

Few students do anything before a lecture, which the book calls unfortunate.

BEFORE

Read the next section, even quickly.

Try two problems from the end of the chapter.

Skim the last set of notes.

Write down one question you want answered.

DURING

Sit where you actually pay attention.

Phone away. Not face down. Away.

Notes by hand. Slower means more selective, and it sticks better.

Say you are lost when you are lost. Others are too.

AFTER

Annotate the notes within a day, while you still remember what the shorthand meant.

Work the assigned problems, then one that was not.

Go over the problems with a study partner.

Explain one idea out loud to somebody else.

Pick one item from one column and do it this week.

What to take from today

A short list, because this lecture has a lot in it.

Bloom. Learning is not only remembering. Courses ask you to apply, analyze, evaluate and create.

80/20. Most learning happens when you use the material after class.

Learning-style scales. Know your preferences, then study on both sides of each scale.

Reinforcement. Repeated exposure over time beats relearning under pressure.

Metacognition. Watch your own learning process and change the method when it is not working.

Pick one change small enough that you will actually do it.